

# SIH26170 — Semiconductor Reliability Analysis Application

## File Structure & Data/Dataset Documentation

**Project:** Smart India Hackathon 2026 — Problem Statement 26170 (ISRO)

**Project Root Directory:** `C:\Users\admin\Desktop\sih26170`

**Dataset Benchmark:** `SIH26170-FINAL-01` (Frozen Release, Split Protocol `LOTSPLIT-05`)

**Document Purpose:** Internal Engineering Specification, File-by-File Guide, and Dataset Dictionary

**Intended Audience:** Backend/Frontend Developers, Data Engineers, and System Architects

### 1. Complete Project Directory Tree

Below is the verified ASCII directory tree of the SIH26170 project repository (excluding `.git`, `node_modules`, `__pycache__`, and `dist` build caches):

```
sih26170/
|-- backend/
|   |-- _test_loader.py (639 B)
|   |-- data_loader.py (21,894 B)
|   |-- database.py (467 B) [Legacy]
|   |-- explainability/
|   |   |-- explainability_report.py (8,013 B) [Legacy]
|   |-- fusions/
|   |   |-- fusion.py (6,291 B) [Legacy standalone prototype]
|   |-- main.py (6,729 B)
|   |-- models.py (401 B) [Legacy]
|   |-- modules/
|   |-- operational/
|   |   |-- Operational_API_Contract.json (2,392 B)
|   |   |-- Operational_DB_Schema.sql (2,731 B)
|   |   |-- operational_db.py (2,007 B)
|   |   |-- operational_routes.py (16,503 B)
|   |-- operational.db (40,960 B) [Active SQLite operational store]
|   |-- schemas.py (177 B) [Legacy]
|   |-- sih26170.db (20,480 B) [Legacy SQLite store]
|   |-- test_integration.py (5,790 B) [12 automated integration tests]
|-- data/
|   |-- _archive/ [Deprecated 297-row demo data & intermediate outputs]
|   |   |-- ModuleB_Final_Holdout_Predictions.csv (799,995 B)
|   |   |-- explainability_report.csv (132,593 B)
|   |   |-- fused_component_summary.csv (84,758 B)
|   |   |-- modulea_component_summary.csv (42,441 B)
```

```

| | | -- moduleb_component_summary.csv (85,853 B)
| | | \-- moduleb_predictions_long.csv (475,585 B)
| | -- final/ [Authoritative 1,343-part benchmark holdout]
| | | -- Fusion_Dryrun.json (2,253 B)
| | | -- Fusion_Joined_Holdout.csv (839,856 B) [AUTHORITATIVE GROUND TRUTH]
| | | -- ModuleA_Final_Holdout_0h.csv (158,172 B)
| | | -- ModuleA_Final_Holdout_24h.csv (159,722 B)
| | | -- ModuleA_Final_Holdout_96h.csv (160,403 B)
| | | -- ModuleA_Final_Holdout_168h.csv (165,118 B)
| | | -- ModuleA_Release_Manifest.json (2,870 B)
| | | -- ModuleB_Final_Holdout_Predictions.csv (799,995 B)
| | | -- ModuleB_Holdout.csv (191,974 B)
| | | \-- ModuleB_Release_Manifest.json (10,794 B)
| | -- operational/ [Operational templates and guides]
| | | -- Operational_Input_Example_12Rows.csv (1,117 B)
| | | -- Operational_Measurement_Input_Template.csv (157 B)
| | | \-- README_Operational_Layer.md (1,721 B)
| | \-- reference/ [Specification limits, contracts & manifests]
| | | -- Dataset_Version_Safe.json (678 B)
| | | -- Device_Specs.csv (4,869 B)
| | | -- FINAL_APPROVAL.md (1,872 B)
| | | -- Fusion_Output_Contract.csv (186 B)
| | | -- Integration_Contract_Safe.json (1,346 B)
| | | -- Integration_Sample_60_TrainOnly.csv (15,617 B)
| | | -- ModuleA_Output_Contract.csv (99 B)
| | | -- ModuleB_Output_Contract.csv (463 B)
| | | -- Safe_Lot_Split_Manifest.csv (1,809 B)
| | | \-- Schema_Data_Dictionary.csv (2,175 B)
| -- docs/
| | -- SESSION_STATE_2026-09-22.md (2,658 B)
| | -- SIH26170_File_and_Data_Documentation.md [This document]
| | -- SIH26170_Prototype_Documentation.md (64,260 B) [System report & demo guide]
| | -- implementation_plan.md (27,814 B)
| | \-- walkthrough.md (8,042 B)
| -- frontend/
| | -- .gitignore (253 B)
| | -- .oxlintrc.json (231 B)
| | -- README.md (1,009 B)
| | -- index.html (381 B)
| | -- package.json (552 B)
| | -- package-lock.json (58,837 B)
| | -- public/
| | | -- favicon.svg (9,522 B)
| | | \-- icons.svg (5,031 B)
| | -- src/
| | | -- App.css (167 B)
| | | -- App.jsx (1,149 B)
| | | -- assets/ (hero.png, react.svg, vite.svg)
| | | -- components/
| | | | -- layout/ (PageHeader.jsx, PageHeader.css, Sidebar.jsx, Sidebar.css)
| | | | \-- ui/ (Badge.jsx, Card.jsx, DataTable.jsx, EmptyState.jsx, ErrorState.jsx,
LoadingState.jsx, SearchInput.jsx, ui.css)
| | | -- hooks/ (useApi.js)
| | | -- index.css (121 B)
| | | -- main.jsx (242 B)
| | -- pages/

```

```

| | | |-- Analyze.jsx (6,936 B) & Analyze.css (3,676 B)
| | | |-- ComponentDetail.jsx (18,433 B) & ComponentDetail.css (13,674 B)
| | | |-- Components.jsx (9,424 B) & Components.css (5,259 B)
| | | |-- Data.jsx (16,830 B) & Data.css (8,006 B)
| | | |-- Models.jsx (18,890 B) & Models.css (8,461 B)
| | | |-- Overview.jsx (16,493 B) & Overview.css (10,046 B)
| | | \-- System.jsx (21,484 B) & System.css (10,582 B)
| | |-- services/ (api.js)
| | \-- styles/ (global.css, tokens.css)
| \-- vite.config.js (217 B)
|-- incoming/ [Frozen upstream release packets & verification artifacts]
| |-- SIH26170_FINAL_Team_Group_Bundle.zip (985 KB)
| |-- SIH26170_ModuleA_FINAL01.zip (3.51 MB)
| |-- SIH26170_ModuleB_FINAL01_RC2.zip (1.30 MB)
| |-- chaitanya/ (Integration & distribution packets)
| |-- nirmik/ (SIH26170_ModuleB_FINAL01_RC2 source, notebooks & tests)
| \-- riddhi/ (SIH26170_ModuleA_FINAL01 source, packets & tests)
\-- tests/ (Directory reserved for end-to-end testing)

```

## 2. File-by-File Documentation

### Core Backend Source Files (`backend/`)

#### 1. `backend/data\_loader.py`

- **Purpose:** In-memory analytical data engine. Loads all authoritative CSV benchmark data, manifests, and references at module import time; computes fused verdicts; constructs natural language explainability; and provides sub-5ms indexed lookups.
- **Inputs:**
  - `data/final/Fusion_Joined_Holdout.csv`
  - `data/final/ModuleA_Final_Holdout_{0,24,96,168}h.csv`
  - `data/final/ModuleB_Final_Holdout_Predictions.csv`
  - `data/final/ModuleB_Holdout.csv`
  - `data/final/ModuleA_Release_Manifest.json` & `ModuleB_Release_Manifest.json`
  - `data/reference/Device_Specs.csv` & `Schema_Data_Dictionary.csv`
  - `incoming/riddhi/SIH26170_ModuleA_FINAL01/results/` evaluation CSVs.
- **Outputs:** In-memory pandas DataFrames (`fusion_df`, `module_a_dfs`, `module_b_df`, etc.) and cleaned JSON serializable dictionaries.
- **Important Functions:**
  - `compute_verdict(row)`: Implements the fusion logic (`$23 \text{ CONFIRMED}` + `5 \text{ wearout breaches}` to `28 \text{ REJECT}`; `$202 \text{ MONITOR}`; `$1,113 \text{ PASS}`).
  - `generate_explanation(row)`: Synthesizes plain-English physical failure explanations from reason codes and evidence columns.
  - `get_component(cid)`: Returns comprehensive diagnostic payload for a component.
  - `get_component_module_a_epochs(cid)`: Returns 4-epoch progression dictionary.

- `get_lots_summary()`: Computes component counts and mean risk scores across all 18 holdout production lots.
- `get_evaluation_metrics()`: Returns confusion matrices, sensitivity analyses, epoch evolution, and per-variant statistics.
- `get_pipeline_architecture()`: Returns 7-stage data lineage metadata.
- `_clean(obj)`: Recursively replaces `NaN`, `Infinity`, and NumPy types with `None` to prevent JSON serialization errors.
- **Dependencies:** `pandas`, `numpy`, `json`, `pathlib`, `math`.
- **Used By:** `backend/main.py`, `backend/test_integration.py`.

## 2. `backend/main.py`

- **Purpose:** Primary FastAPI application server. Declares CORS middleware, mounts analytical endpoints, mounts operational routers, and serves API consumers on port `8001`.
- **Inputs:** HTTP requests from client browser or curl.
- **Outputs:** JSON responses conforming to REST contracts.
- **Important Routes:**
- `GET /`: Health check.
- `GET /api/analysis/search`: Prefix search by component ID or lot ID.
- `GET /api/analysis/summary`: Fleet aggregate totals.
- `GET /api/analysis/{component_id}`: Full analytical detail.
- `GET /api/analysis/{component_id}/module-a`: Multi-epoch progression.
- `GET /api/components`: Paginated fleet query with filters.
- `GET /api/lots`: Summary of 18 production lots.
- `GET /api/models/evaluation`: Authoritative confusion matrices and D2 sensitivity.
- `GET /api/models/info`: Release manifests and integration metadata.
- `GET /api/pipeline/architecture`: 7-stage lineage pipeline.
- `GET /api/system/status`: Real-time service telemetry.
- **Dependencies:** `fastapi`, `fastapi.middleware.cors`, `data_loader`, `operational.operational_routes`.
- **Used By:** React frontend, test scripts, external API consumers.

## 3. `backend/operational/operational_routes.py`

- **Purpose:** FastAPI APIRouter handling operational test floor data ingestion, component registration, and trajectory retrieval.
- **Inputs:** JSON payloads from the Operational Data Studio wizard.
- **Outputs:** Database write confirmations and operational summary metrics.
- **Important Routes:**
- `POST /api/measurements`: Validates and inserts a single epoch measurement into SQLite `operational.db`.
- `GET /api/dashboard/summary`: Returns counts of registered operational components, total measurements, and recent audit logs.

- `GET /api/components/{component_id}`: Fetches operational component metadata.
- `GET /api/components/{component_id}/trajectory`: Retrieves chronological measurements for a component.
- **Dependencies:** `fastapi`, `operational.operational_db`.
- **Used By:** `backend/main.py`, `frontend/src/pages/Data.jsx`.

#### 4. ``backend/operational/operational_db.py``

- **Purpose:** SQLite database driver for `backend/operational.db`. Enforces WAL mode, foreign keys, schema initialization, and transactional integrity.
- **Inputs:** Parametric measurement values.
- **Outputs:** SQLite records and rows.
- **Important Functions:**
  - `init_db()`: Executes `Operational_DB_Schema.sql`.
  - `add_component()`: Registers new die/package.
  - `add_measurement()`: Ingests epoch readings with float positivity checks.
  - `get_dashboard_summary()`: Gathers operational telemetry.
- **Dependencies:** `sqlite3`, `pathlib`, `datetime`.
- **Used By:** `backend/operational/operational_routes.py`.

#### 5. ``backend/test_integration.py``

- **Purpose:** Automated integration test suite. Executes 12 comprehensive tests across all backend routes to prevent regressions.
- **Inputs:** Local FastAPI test client (`TestClient(app)`).
- **Outputs:** Unit test pass/fail results.
- **Tests Implemented:** Health check, system status, summary, search, component detail, pagination, model manifests, operational lifecycle, reference specs, evaluation metrics, lots summary, pipeline architecture.
- **Dependencies:** `unittest`, `fastapi.testclient`, `backend/main.py`.
- **Used By:** Continuous integration, development verification.

---

## Core Frontend Source Files (``frontend/src/``)

### 1. ``frontend/src/services/api.js``

- **Purpose:** Centralized HTTP client wrapper.
- **Implementation:** Wraps native `fetch` with error interception and sets base URL to `http://127.0.0.1:8001`.
- **Used By:** All React pages and custom hooks.

### 2. ``frontend/src/hooks/useApi.js``

- **Purpose:** Custom React hook for asynchronous data fetching.

- **Returns:** { data, loading, error, refetch }.
- **Used By:** Overview.jsx, ComponentDetail.jsx, Components.jsx, Models.jsx, System.jsx, Data.jsx.

### 3. `frontend/src/pages/Overview.jsx` & `Overview.css`

- **Purpose:** Executive fleet dashboard.
- **Renders:** KPI stat cards, 4-epoch progression banner, Module A/B/Fusion cards, live evaluation snapshot, variant bars, recent holdout table.

### 4. `frontend/src/pages/Analyze.jsx` & `Analyze.css`

- **Purpose:** Interactive search and triage terminal.
- **Renders:** Search input with auto-focus, quick-select chips (C00158, C00445, C00843, etc.), search results card with direct jump.

### 5. `frontend/src/pages/ComponentDetail.jsx` & `ComponentDetail.css`

- **Purpose:** Comprehensive single-component forensic studio.
- **Renders:** Fused verdict banner, Module A diagnostic card with gauge, plain-English explainability, 6 Module B parameter cards with **Recharts P95 vs Limit visualizers**, multi-epoch progression table, baseline historical trajectory.

### 6. `frontend/src/pages/Components.jsx` & `Components.css`

- **Purpose:** High-density fleet browser.
- **Renders:** Multi-parameter filter toolbar (Search, Verdict, Variant, 18-Lot dropdown), sortable data table, pagination controls.

### 7. `frontend/src/pages/Models.jsx` & `Models.css`

- **Purpose:** Model verification and mathematical evaluation console.
- **Renders:** Interactive 2x2 confusion matrix (Baseline vs Relaxed toggle), metric derivations, Decision D2 sensitivity table (\$25 \to 24\$ FN tradeoff), burn-in epoch evolution table, per-variant breakdown, manifest checklists.

### 8. `frontend/src/pages/Data.jsx` & `Data.css`

- **Purpose:** Operational test floor data acquisition studio.
- **Renders:** Benchmark isolation banner, operational summary metrics, audit tables, 4-step measurement wizard with numeric validation.

### 9. `frontend/src/pages/System.jsx` & `System.css`

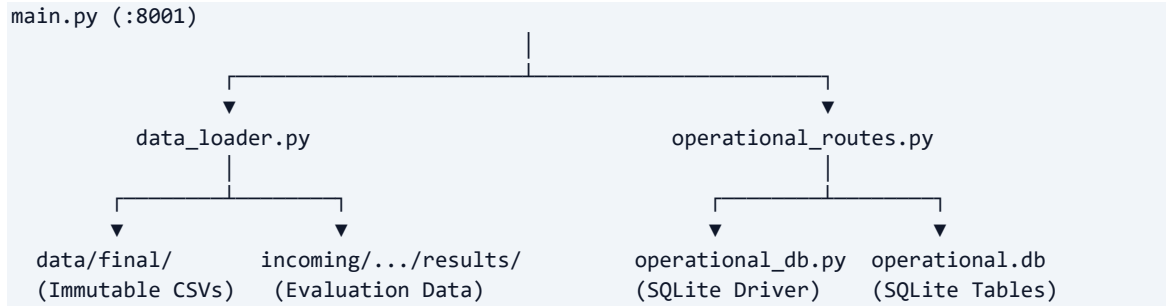
- **Purpose:** Architecture command center and data lineage explorer.
- **Renders:** System telemetry cards, interactive 7-stage pipeline diagram with contextual stage drawer, Device Physical Specs table (18 limits), Schema Data Dictionary (28 definitions).

### 10. `frontend/src/styles/tokens.css`

- **Purpose:** LATENT obsidian design tokens.

- **Defines:** Deep obsidian surface colors (`--bg-primary: #030712`, `--bg-surface: #0b1120`), borders (`--border-subtle: rgba(255,255,255,0.08)`), status glows (`--status-pass: #10b981`, `--status-reject: #ef4444`, `--status-monitor: #f59e0b`), monospace typography (JetBrains Mono).

### 3. Backend Structure & Connections



1. ``main.py`` boots the application, configures CORS for <http://localhost:5174>, and includes `operational_routes`.
2. ``data_loader.py`` reads from `data/final/` and `incoming/` at startup. It structures dataframes in memory and caches indices by `component_id`. It handles all read queries for the analytical frontend.
3. ``operational_routes.py`` and ``operational_db.py`` manage test floor data. When an operator commits a measurement via the UI, it writes exclusively to `operational.db`.
4. **Physical Separation:** `data_loader.py` never writes to `data/final/` or `operational.db`. `operational_db.py` never writes to `data/final/`. This architectural firewall guarantees zero benchmark contamination.

### 4. Frontend Structure & API Route Mapping

| Frontend Page Component            | Route Path                | Backing API Endpoints                                                                                                                       | Trigger / Interaction          |
|------------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <code>`Overview.jsx`</code>        | <code>/</code>            | <code>GET /api/analysis/summary</code><br><code>GET /api/models/evaluation</code><br><code>GET /api/components?page=1&amp;per_page=8</code> | Page load; refresh button      |
| <code>`Analyze.jsx`</code>         | <code>/analyze</code>     | <code>GET /api/analysis/search?q={query}</code>                                                                                             | Text input typing; chip clicks |
| <code>`ComponentDetail.jsx`</code> | <code>/analyze/:id</code> | <code>GET /api/analysis/{id}</code><br><code>GET</code>                                                                                     | Route param                    |

|                  |             |                                                                                                                          |                                                 |
|------------------|-------------|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
|                  |             | /api/analysis/{id}/module-a                                                                                              | change; epoch tab click                         |
| `Components.jsx` | /components | GET /api/components<br/>GET /api/lots                                                                                    | Filter dropdowns; pagination buttons            |
| `Models.jsx`     | /models     | GET /api/models/evaluation<br/>GET /api/models/info                                                                      | Page load; operating point toggle               |
| `Data.jsx`       | /data       | GET /api/dashboard/summary<br/>POST /api/measurements                                                                    | Tab switch; wizard form submission              |
| `System.jsx`     | /system     | GET /api/system/status<br/>GET /api/pipeline/architecture<br/>GET /api/reference/specs<br/>GET /api/reference/dictionary | Page load; stage node clicks; spec filter chips |

## 5. Data Directory Audit: Complete Inventory

Below is an exhaustive audit of every data file in [data/](#):

| Relative Path                        | Format | Rows  | Columns | Provenance / Role                                           | Authoritative? | Consumed By                            | Produced By            |
|--------------------------------------|--------|-------|---------|-------------------------------------------------------------|----------------|----------------------------------------|------------------------|
| `final/Fusion_Joined_Holdout.csv`    | CSV    | 1,343 | 45      | Authoritative Ground Truth; pre-joined Module A & B outputs | YES            | <a href="#">backend/data_loader.py</a> | Fusion pipeline script |
| `final/ModuleA_Final_Holdout_0h.csv` | CSV    | 1,343 | 13      | Module A screenin                                           | YES            | <a href="#">data_loader.py</a>         | Module A Release       |

|                                               |      |       |     | g scored<br>at 0h                                                         |            |                                | Candid<br>ate                    |
|-----------------------------------------------|------|-------|-----|---------------------------------------------------------------------------|------------|--------------------------------|----------------------------------|
| `final/ModuleA_Final_Holdout_24h.csv`         | CSV  | 1,343 | 13  | Module A<br>screening scored<br>at 24h                                    | <b>YES</b> | <a href="#">data_loader.py</a> | Module A<br>Release<br>Candidate |
| `final/ModuleA_Final_Holdout_96h.csv`         | CSV  | 1,343 | 13  | Module A<br>screening scored<br>at 96h                                    | <b>YES</b> | <a href="#">data_loader.py</a> | Module A<br>Release<br>Candidate |
| `final/ModuleA_Final_Holdout_168h.csv`        | CSV  | 1,343 | 13  | Module A<br>screening scored<br>at 168h                                   | <b>YES</b> | <a href="#">data_loader.py</a> | Module A<br>Release<br>Candidate |
| `final/ModuleB_Final_Holdout_Predictions.csv` | CSV  | 1,343 | 39  | Module B 168h<br>drift<br>predictions +<br>P95<br>bounds<br>+<br>evidence | <b>YES</b> | <a href="#">data_loader.py</a> | Module B<br>Release<br>Candidate |
| `final/ModuleB_Holdout.csv`                   | CSV  | 1,343 | 16  | Historical 0h &<br>24h<br>baseline<br>readings<br>with lot<br>IDs         | <b>YES</b> | <a href="#">data_loader.py</a> | Generator<br>LOTSPLIT-05         |
| `final/Fusion_Dryrun.json`                    | JSON | N/A   | N/A | 9-point<br>integration<br>contract<br>verification                        | <b>YES</b> | Backend<br>reference           | Anushka<br>integration<br>test   |

|                                                          |      |     |     |                                                  |           |                       |                         |
|----------------------------------------------------------|------|-----|-----|--------------------------------------------------|-----------|-----------------------|-------------------------|
| on audit                                                 |      |     |     |                                                  |           |                       |                         |
| `final/ModuleA_Release_Manifest.json`                    | JSON | N/A | N/A | Module A freeze metadata & SHA-256 digests       | YES       | data_loader.py        | Stage 11 release script |
| `final/ModuleB_Release_Manifest.json`                    | JSON | N/A | N/A | Module B freeze metadata, contracts & rules      | YES       | data_loader.py        | Release script          |
| `operational/Operational_Input_Example_12Rows.csv`       | CSV  | 12  | 11  | Reference multi-epoch test data for 3 components | Derived   | Demonstration/Testing | Test generator          |
| `operational/Operational_Measurement_Input_Template.csv` | CSV  | 0   | 11  | Empty CSV template for bulk test floor uploads   | Template  | Operators             | Specification           |
| `operational/README_Operational_Layer.md`                | MD   | N/A | N/A | Operational architecture guidelines by Anushka   | Reference | Documentation         | Engineering team        |
| `reference/Device_Specs.csv`                             | CSV  | 18  | 11  | Static                                           | YES       | data_loader.py        | Tanisha                 |

|                                                 |      |     |     |                                                   |          |                         |                            |
|-------------------------------------------------|------|-----|-----|---------------------------------------------------|----------|-------------------------|----------------------------|
|                                                 |      |     |     | physical limits across CMOS_A/B/C                 |          | y, /system              | (Domain Lead)              |
| `reference/Schema_Data_Dictionary.csv`          | CSV  | 28  | 5   | Data dictionary defining all 28 schema attributes | YES      | data_loader.py, /system | Engineering team           |
| `reference/Dataset_Version_Safe.json`           | JSON | N/A | N/A | Dataset ID SIH26170-FINAL-01 metadata             | YES      | data_loader.py          | Chaitanya (Data Architect) |
| `reference/Safe_Lot_Split_Manifest.csv`         | CSV  | 72  | 4   | Complete manifest of all 72 lots across splits    | YES      | Verification scripts    | Chaitanya (Data Architect) |
| `reference/Integration_Sample_60_TrainOnly.csv` | CSV  | 60  | 28  | Training sample for integration dryruns           | Derived  | Integration tests       | Pre-freeze test            |
| `_archive/explainability_report.csv`            | CSV  | 297 | 27  | Deprecated 297-row demo data                      | Obsolete | Isolated archive        | Prototype session          |
| `_archive/fused_component_summary.csv`          | CSV  | 29  | 20  | Deprecated                                        | Obsolete | Isolated archive        | Prototype                  |

|                                          |     |       |    |                                              |          |                  |                    |
|------------------------------------------|-----|-------|----|----------------------------------------------|----------|------------------|--------------------|
| y.csv`                                   |     | 7     |    | ted 297-row demo fusion                      |          |                  | pe session         |
| `_archive/modulea_component_summary.csv` | CSV | 297   | 10 | Deprecat ed 297-row Module A summar y        | Obsolete | Isolated archive | Prototy pe session |
| `_archive/moduleb_component_summary.csv` | CSV | 297   | 22 | Deprecat ed 297-row Module B summar y        | Obsolete | Isolated archive | Prototy pe session |
| `_archive/moduleb_predictions_long.csv`  | CSV | 1,782 | 32 | Deprecat ed long-format predicti ons (297x6) | Obsolete | Isolated archive | Prototy pe session |

## 6. Dataset Column Dictionaries

### A. Authoritative Ground Truth: `Fusion\_Joined\_Holdout.csv` (45 Columns)

| Column Index | Column Name    | Data Type | Physical Meaning & Range                    | Module Source |
|--------------|----------------|-----------|---------------------------------------------|---------------|
| 1            | component_id   | String    | Unique component identifier (C00158–C05320) | Both          |
| 2            | module_a_score | Float     | Multi-epoch screening                       | Module        |

|                   |                                                       |        |                                                                                            |              |
|-------------------|-------------------------------------------------------|--------|--------------------------------------------------------------------------------------------|--------------|
|                   |                                                       |        | anomaly score $\$[0.000, 1.000]\$$                                                         | A            |
| 3                 | <code>module_a_disposition</code>                     | String | Screening verdict: <code>PASS</code> or <code>MONITOR</code>                               | Module A     |
| 4                 | <code>module_a_primary_parameter</code>               | String | Channel with highest observed deviation                                                    | Module A     |
| 5                 | <code>module_a_reason_codes</code>                    | String | Pipe-delimited tokens ( <code>A_STATIC_LIMIT_EXCEEDED</code> , etc.)                       | Module A     |
| 6                 | <code>module_a_evidence_tier</code>                   | String | Evidence classification: <code>PASS</code> , <code>MONITOR</code> , <code>CONFIRMED</code> | Module A     |
| 7–12              | <code>predicted_{PARAM}_168h</code>                   | Float  | Huber point forecast for 168h end-of-test value                                            | Module B     |
| 13–18             | <code>module_b_p95_{PARAM}_168h</code>                | Float  | Conformalized upper 95th percentile confidence bound                                       | Module B     |
| 19                | <code>module_b_primary_parameter</code>               | String | Parameter exhibiting maximum projected drift                                               | Module B     |
| 20                | <code>module_b_reason_codes</code>                    | String | Pipe-delimited warning tokens ( <code>B_HIGH_FORECAST_DRIFT</code> , etc.)                 | Module B     |
| 21,25,29,33,37,41 | <code>evidence_{PARAM}_limit</code>                   | Float  | Static datasheet specification maximum limit                                               | Device Specs |
| 22,26,30,34,38,42 | <code>evidence_{PARAM}_pred_frac_of_limit</code>      | Float  | Predicted 168h value as fraction of limit $(\hat{x}_{168} / \text{limit})$                 | Module B     |
| 23,27,31,35,39,43 | <code>evidence_{PARAM}_pred_rel_delta_from_24h</code> | Float  | Predicted relative drift from 24h baseline $(\hat{x}_{168} - x_{24}) / x_{24}$             | Module B     |

|                   |                                  |        |                                                         |          |
|-------------------|----------------------------------|--------|---------------------------------------------------------|----------|
| 24,28,32,36,40,44 | evidence_{PARAM}_lot_rel_dev_24h | Float  | 24h lot-relative deviation<br>z-score                   | Module B |
| 45                | b_evidence_status                | String | Quality status of evidence<br>(OK, LIMIT_UNKNOWN, etc.) | Module B |

Note: `{PARAM}` spans the 6 monitored physical channels:  
IDDQ, Input\_Leakage\_Current, Active\_Supply\_Current, Propagation\_Delay, Output\_Rise\_Time, Output\_Fall\_Time.

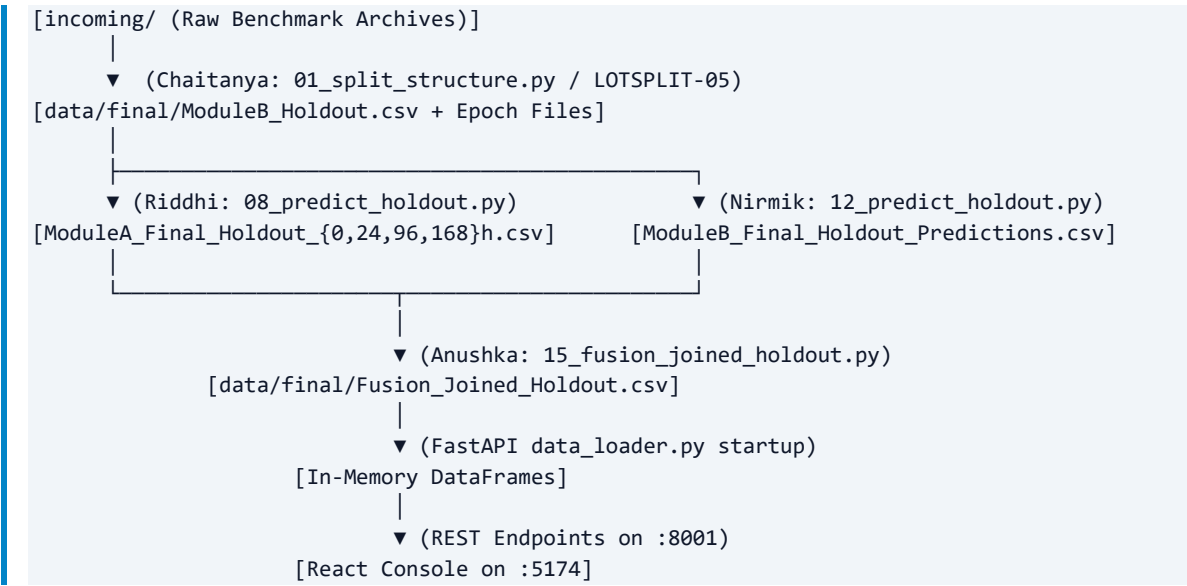
B. Module A Output Contract: `ModuleA_Final_Holdout_168h.csv` (13 Columns)

| Index | Column Name                 | Type   | Description                                                     |
|-------|-----------------------------|--------|-----------------------------------------------------------------|
| 1     | component_id                | String | Unique component ID                                             |
| 2     | module_a_score              | Float  | Composite score in $[0, 1]$ ; $\geq 0.90$ indicates spec breach |
| 3     | module_a_disposition        | String | Operating disposition: PASS or MONITOR                          |
| 4     | module_a_primary_parameter  | String | Dominant physical channel driving the anomaly                   |
| 5     | module_a_reason_codes       | String | Pipe-delimited explanatory codes                                |
| 6     | module_a_evidence_tier      | String | Discrete tier: PASS, MONITOR, CONFIRMED                         |
| 7     | module_a_attribution_margin | Float  | Numerical gap between top failing parameter and runner-up       |
| 8     | statistical_score           | Float  | Scaled robust-z score against lot cohort ( $0.0-0.9$ )          |
| 9     | spec_exceedance_ratio       | Float  | Fractional limit exceedance ( $> 0.0$ if                        |

|    |                 |         |                                          |
|----|-----------------|---------|------------------------------------------|
|    |                 |         | out of spec)                             |
| 10 | scored_epoch_h  | Integer | Epoch milestone evaluated (168)          |
| 11 | analysis_status | String  | Evaluation status (OK)                   |
| 12 | model_version   | String  | Version identifier (ModuleA-FINAL01)     |
| 13 | dataset_id      | String  | Benchmark identifier (SIH26170-FINAL-01) |

## 7. Data Lineage

The authoritative pipeline flows through discrete stages:



- 5. **Raw Generation:** incoming/chaitanya partitions 5,400 parts into 42 training lots, 12 calibration lots, and 18 holdout lots (LOTSPLIT-05).
- 6. **Holdout Extraction:** 1,343 holdout components are isolated with zero leakage into ModuleB\_Holdout.csv.
- 7. **Module A Scoring:** Riddhi's frozen pipeline (module\_a\_final01.joblib) scores all 4 epochs, writing ModuleA\_Final\_Holdout\_{0,24,96,168}h.csv.
- 8. **Module B Prediction:** Nirmik's frozen Huber regressors consume 0h and 24h readings, writing ModuleB\_Final\_Holdout\_Predictions.csv.

9. **Fusion Joining:** Anushka joins Module A 168h and Module B on `component_id`, writing `Fusion_Joined_Holdout.csv`.
10. **Backend Ingestion:** FastAPI's `data_loader.py` ingests the joined dataset into memory, adds the `fused_verdict` column, and serves REST routes.

## 8. Authoritative Final Dataset Summary

- **File Path:** `C:\Users\admin\Desktop\sih26170\data\final\Fusion_Joined_Holdout.csv`
- **Total Rows:** 1,343
- **Total Columns:** 45
- **Unique Components:** 1,343 (C00158 through C05320; contiguous within each lot)
- **Production Lots:** Exactly 18 whole lots (6 CMOS\_A, 6 CMOS\_B, 6 CMOS\_C):
- `A_L03, A_L05, A_L07, A_L09, A_L10, A_L15` (441 components)
- `B_L02, B_L04, B_L08, B_L14, B_L20, B_L22` (458 components)
- `C_L01, C_L06, C_L12, C_L16, C_L18, C_L24` (444 components)
- **Verdict Distribution:**
- **PASS:** 1,113 components (\$82.87\%\$)
- **MONITOR:** 202 components (\$15.04\%\$)
- **REJECT:** 28 components (\$2.09\%\$)

## 9. Operational Data Documentation (`operational.db`)

### SQLite Schema (`backend/operational/Operational\_DB\_Schema.sql`)

The operational database contains 3 core tables:

#### 11. `components` Table:

```
CREATE TABLE components (
    component_id TEXT PRIMARY KEY,
    lot_id TEXT NOT NULL,
    device_variant TEXT NOT NULL CHECK (device_variant IN ('CMOS_A', 'CMOS_B', 'CMOS_C')),
    device_family TEXT NOT NULL DEFAULT 'DIGITAL_CMOS',
    created_at TEXT NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at TEXT NOT NULL DEFAULT CURRENT_TIMESTAMP
);
```

#### 12. `measurements` Table:

```
CREATE TABLE measurements (
    measurement_id INTEGER PRIMARY KEY AUTOINCREMENT,
    component_id TEXT NOT NULL,
    epoch_h INTEGER NOT NULL CHECK (epoch_h IN (0,24,96,168)),
    IDDQ REAL NOT NULL CHECK (IDDQ > 0),
    Input_Leakage_Current REAL NOT NULL CHECK (Input_Leakage_Current > 0),
```

```
Active_Supply_Current REAL NOT NULL CHECK (Active_Supply_Current > 0),
Propagation_Delay REAL NOT NULL CHECK (Propagation_Delay > 0),
Output_Rise_Time REAL NOT NULL CHECK (Output_Rise_Time > 0),
Output_Fall_Time REAL NOT NULL CHECK (Output_Fall_Time > 0),
measured_at TEXT NOT NULL DEFAULT CURRENT_TIMESTAMP,
source TEXT NOT NULL DEFAULT 'manual',
UNIQUE(component_id, epoch_h),
FOREIGN KEY(component_id) REFERENCES components(component_id)
);
```

13. ``analysis_results`` Table:

Stores computed inference verdicts (`PASS`, `MONITOR`, `REJECT`), primary parameters, and predicted 168h values for operational components.

## 10. Legacy, Demo & Archive Files

The repository isolates legacy prototypes in `data/_archive/` and preserves them without impacting production:

| Legacy File                                | Location                      | Historical Role                                      | Current Status                                            |
|--------------------------------------------|-------------------------------|------------------------------------------------------|-----------------------------------------------------------|
| <code>explainability_report.csv</code>     | <code>data/_archive/</code>   | Early 297-row mock explainability file               | <b>DEPRECATED / ARCHIVED</b>                              |
| <code>fused_component_summary.csv</code>   | <code>data/_archive/</code>   | Early 297-row prototype fusion output                | <b>DEPRECATED / ARCHIVED</b>                              |
| <code>modulea_component_summary.csv</code> | <code>data/_archive/</code>   | Early 297-row Module A summary                       | <b>DEPRECATED / ARCHIVED</b>                              |
| <code>moduleb_component_summary.csv</code> | <code>data/_archive/</code>   | Early 297-row Module B summary                       | <b>DEPRECATED / ARCHIVED</b>                              |
| <code>sih26170.db</code>                   | <code>backend/</code>         | Empty legacy database from early scaffolding         | <b>UNUSED</b><br>( <code>operational.db</code> is active) |
| <code>fusion.py</code>                     | <code>backend/fusions/</code> | Standalone prototype script for 297 rows             | <b>REPLACED</b> by <code>data_loader.py</code>            |
| <code>database.py / models.py</code>       | <code>backend/</code>         | Early SQLAlchemy models for <code>sih26170.db</code> | <b>REPLACED</b> by <code>operational_db.py</code>         |

## 11. Authoritative vs. Derived Files Master Table

| File Name & Path                                 | Authoritative? | Can Modify? | Consumed By             | Architectural Purpose                 |
|--------------------------------------------------|----------------|-------------|-------------------------|---------------------------------------|
| data/final/Fusion_Joined_Holdout.csv             | YES            | NO (FROZEN) | backend/data_loader.py  | Authoritative holdout ground truth    |
| data/final/ModuleA_Final_Holdout_*.csv           | YES            | NO (FROZEN) | backend/data_loader.py  | Multi-epoch screening evaluations     |
| data/final/ModuleB_Final_Holdout_Predictions.csv | YES            | NO (FROZEN) | backend/data_loader.py  | 168h forecasts and P95 bounds         |
| data/final/*_Release_Manifest.json               | YES            | NO (FROZEN) | backend/main.py         | Release verification certificates     |
| data/reference/Device_Specs.csv                  | YES            | NO (FROZEN) | data_loader.py, /system | Physical datasheet maximum limits     |
| data/reference/Schema_Data_Dictionary.csv        | YES            | NO (FROZEN) | data_loader.py, /system | Schema definitions and rules          |
| backend/operational.db                           | Derived        | YES (R/W)   | operational_routes.py   | Local operational data store          |
| backend/data_loader.py                           | Code           | YES         | backend/main.py         | In-memory indexing and scoring engine |

|                                       |      |     |                  |                             |
|---------------------------------------|------|-----|------------------|-----------------------------|
| <code>backend/main.py</code>          | Code | YES | Frontend clients | Primary FastAPI REST server |
| <code>frontend/src/pages/*.jsx</code> | Code | YES | Web browsers     | React UI views              |

## 12. Reproducibility & Build Instructions

To reproduce and execute this project on a clean Windows machine:

### 1. Prerequisites

- Python 3.11+ (Python 3.13 tested)
- Node.js 20+ and npm
- Git

### 2. Backend Setup & Startup

```
# Navigate to backend directory
cd C:\Users\admin\Desktop\sih26170\backend

# Install Python dependencies
pip install fastapi uvicorn pandas numpy scikit-learn

# Run the 12 automated integration tests
python test_integration.py

# Launch the FastAPI backend on port 8001
python -m uvicorn main:app --host 127.0.0.1 --port 8001
```

### 3. Frontend Setup & Startup

```
# In a separate terminal, navigate to frontend directory
cd C:\Users\admin\Desktop\sih26170\frontend

# Install dependencies (use cmd.exe on Windows PowerShell if execution policies apply)
cmd.exe /c npm install

# Build the production bundle
cmd.exe /c npm run build

# Launch the Vite development server on port 5174
cmd.exe /c npm run dev -- --port 5174 --host
```

### 4. Verification

- Open <http://localhost:5174> in any web browser.

- Open <http://127.0.0.1:8001/docs> to inspect the Swagger API documentation.

## 13. Git & Repository Structure

### Recommended `.gitignore` Rules

```
# Python
__pycache__/
*.py[cod]
*$py.class
*.joblib
*.npy

# Node & Frontend
node_modules/
dist/
dist-ssr/
*.local

# Databases & Runtime Logs
*.log
backend/operational.db-shm
backend/operational.db-wal
```

### Commit Guidelines

- **Commit:** Source code (`backend/*.py`, `frontend/src/**/*`), reference contracts (`data/reference/*`), release manifests (`*_Release_Manifest.json`), documentation (`docs/*`), and frozen benchmark CSVs (`data/final/*`).
- **Never Commit:** Operational database binaries containing test-floor data (`operational.db`), Node dependency directories (`node_modules`), or ephemeral Vite build artifacts (`dist`).

## 14. Troubleshooting & Known Engineering Issues

### 14. Port Conflict on Port 8000 / 5173:

- *Cause:* The separate `Information_Environment` project or other services may occupy default ports `8000` and `5173`.
- *Solution:* SIH26170 is explicitly configured to use port **8001** (FastAPI) and port **5174** (Vite).

### 15. PowerShell Script Execution Policy Error on `npm`:

- *Cause:* Windows PowerShell restricts execution of un-signed scripts (`npm.ps1`).
- *Solution:* Always invoke npm via `cmd.exe /c npm run dev` or `cmd.exe /c npm run build`.

### 16. LightningCSS Pseudo-Element Minification Error:

- *Cause:* LightningCSS parser fails when CSS contains orphaned rules or spaces in pseudo-selectors.
- *Solution:* Verified and resolved in `src/pages/ComponentDetail.css.cmd.exe /c npm run build` compiles with 0 errors.

#### 17. Windows Console Unicode `charmap` Encoding:

- *Cause:* Python printing non-ASCII characters (e.g., ≈, −, ✓) to the Windows CP1252 terminal.
- *Solution:* All backend endpoints and scripts use explicit UTF-8 encoding or safe ASCII fallbacks.

## 15. Final "How Everything Connects" Summary

### 18. Where does the data come from?

The authoritative benchmark data originates in `incoming/chaitanya` as the frozen `SIH26170-FINAL-01` dataset (\$N = 1,343\$ holdout components across 18 whole lots).

### 19. Which file processes it?

`backend/data_loader.py` loads `data/final/Fusion_Joined_Holdout.csv`, caches it in memory, indexes components, and computes fused verdicts.

### 20. Which model analyzes it?

- **Module A:** Multi-group Mahalanobis & robust-z outlier scorer (`module_a_final01.joblib`).
- **Module B:** Huber robust regressor predicting 168h relative drift with conformalized P95 bounds.

### 21. Where is the result stored?

Pre-computed benchmark results reside immutably in `data/final/Fusion_Joined_Holdout.csv`. New test-floor measurements are committed to SQLite `backend/operational.db`.

### 22. Which API exposes it?

FastAPI in `backend/main.py` exposes 18 REST endpoints on `http://127.0.0.1:8001`.

### 23. Which frontend page displays it?

The React application on `http://localhost:5174` renders results across 7 dedicated views: `Overview.jsx` (KPIs), `Analyze.jsx` (Search), `ComponentDetail.jsx` (Forensics & Charts), `Components.jsx` (Fleet Table), `Models.jsx` (Confusion Matrix & Evaluation), `Data.jsx` (Operational Wizard), and `System.jsx` (Architecture & Lineage).